

Exp no :08

Date:

Use Steg-Expose to Detect Hidden Data in Images

Description

StegExpose is a specialized tool used for **steganography analysis**. It works by evaluating the **statistical properties** of an image to estimate the probability of hidden data being embedded within it. This process helps forensic investigators detect steganography techniques like Least Significant Bit (LSB) embedding.

Prerequisites

- Java Runtime Environment (JRE): StegExpose is a Java-based application and requires JRE to run.
- StegExpose Tool: Download the latest .jar file from the official StegExpose GitHub repository.

Step-by-Step Process

1. Download and Set Up StegExpose

1. Download the tool: Obtain the StegExpose.jar file from the GitHub repository.
2. Install Java: Ensure JRE is installed on your machine.
3. Prepare environment: Place the StegExpose.jar file in a dedicated working folder.

```
Microsoft Windows [Version 10.0.26200.9278]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Sarvegna>java -version
java version "26.0.2.1" 2026-08-18
Java(TM) SE Runtime Environment (build 26.0.2.1+1-7)
Java HotSpot(TM) 64-Bit Server VM (build 26.0.2.1+1-7, mixed mode, sharing)

C:\Users\Sarvegna>
```

2. Select Images for Analysis

- Collect the images you suspect might contain hidden data.
- StegExpose supports common image formats such as .png, .jpg, and .bmp.

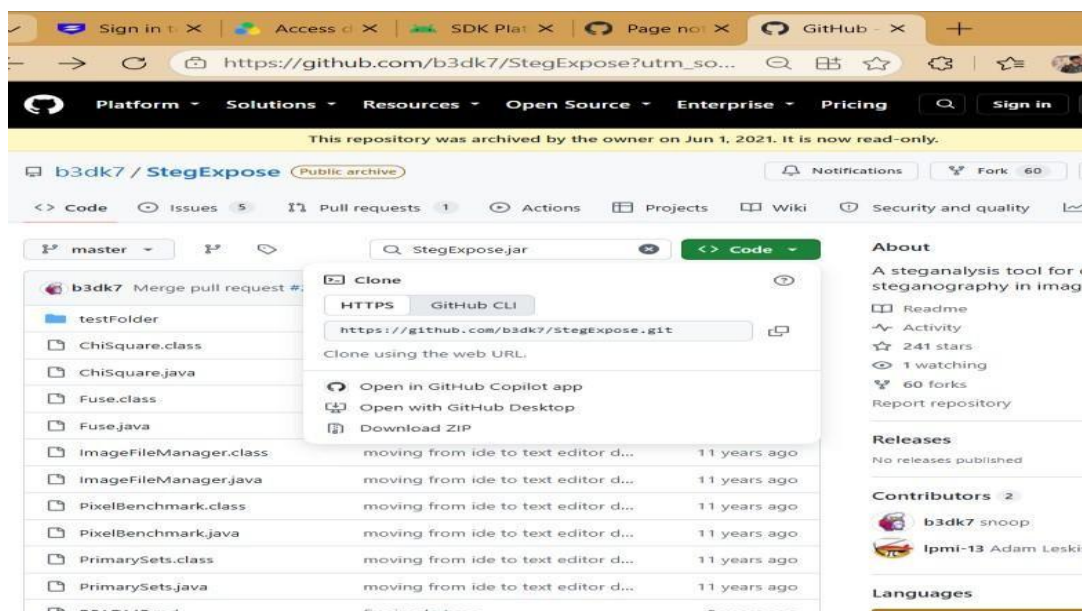
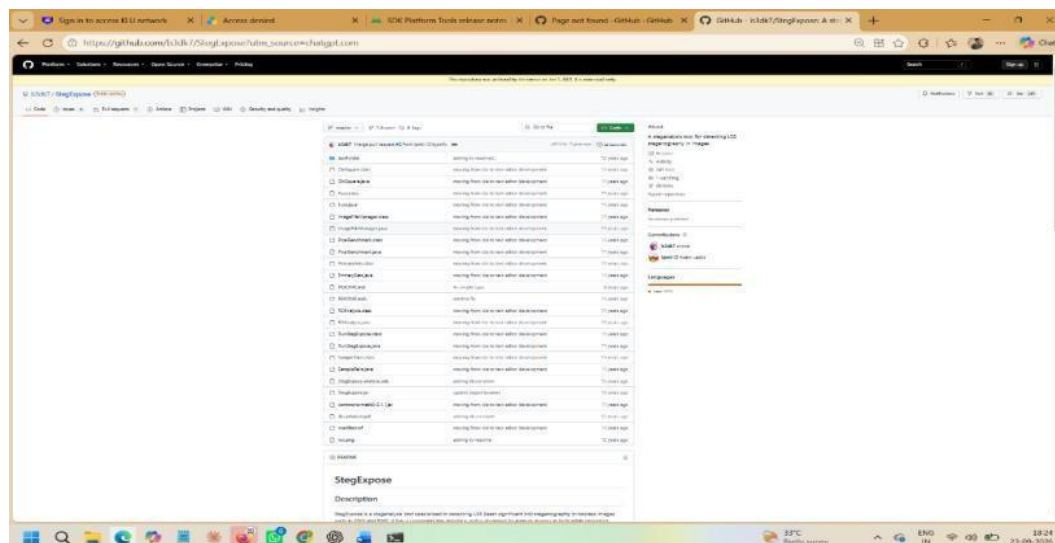
```
C:\Windows\System32\cmd.e X + v

Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Sarvegna\Downloads\platform-tools-latest-windows\platform-tools>adb version
Android Debug Bridge version 1.0.41
Version 37.0.1-15733141
Installed as C:\Users\Sarvegna\Downloads\platform-tools-latest-windows\platform-tools\adb.exe
Running on Windows 10.0.26200
```

3. Open Command Line or Terminal

- Navigate to the folder where the StegExpose.jar file is located using your Command Prompt (Windows) or Terminal (Linux/macOS).



3. Run StegExpose on an Image

- Use the following command structure to analyze a single image for hidden data:

Command Structure:

```
java -jar StegExpose.jar <image_file_path>
```

Example:

```
java -jar StegExpose.jar test_image.png
```

4. Analyze the Output

- StegExpose calculates a **"suspect" score** ranging from 0 to 1. **The higher the score, the more likely steganography is present.**

Score Range	Interpretation (Suggested Thresholds)
Less than 0.2	Image is considered clean (no hidden data detected).
0.2 - 0.3	Possibly some hidden data is present.
Above 0.3	Likely that steganography is present.

- **Example Output Analysis:**

```
java -jar StegExpose.jar suspect_image.png
```

Analyzing suspect_image.png...

Result: 0.4

Steganography likely present

```
C:\Users\Sarvegna\Downloads\StegExpose-master\StegExpose-master>java -jar StegExpose.jar testFolder default default steganalysisOfTestFolder

C:\Users\Sarvegna\Downloads\StegExpose-master\StegExpose-master>dir steganalysisOfTestFolder\*
Volume in drive C is Windows-SSD
Volume Serial Number is DC08-0AC2

Directory of C:\Users\Sarvegna\Downloads\StegExpose-master\StegExpose-master

23-09-2026  18:40                2,255 steganalysisOfTestFolder
             1 File(s)                2,255 bytes
             0 Dir(s) 283,839,115,264 bytes free

C:\Users\Sarvegna\Downloads\StegExpose-master\StegExpose-master>
```

```
C:\Windows\System32\cmd.exe
C:\Users\Sarvegna>cd C:\Users\Sarvegna\Downloads\StegExpose-master\StegExpose-master
C:\Users\Sarvegna\Downloads\StegExpose-master\StegExpose-master>dir
Volume in drive C is Windows-SSD
Volume Serial Number is DC8D-8AC2

Directory of C:\Users\Sarvegna\Downloads\StegExpose-master\StegExpose-master

23-09-2026 18:40 <DIR> .
23-09-2026 18:40 <DIR> ..
23-09-2026 18:40          4,608 Chisquare.class
23-09-2026 18:38          128 Chisquare.java
23-09-2026 18:38    1,590,125 diamond-research1.1.1.jar
23-09-2026 18:38    681,660 dianoretrrion.pdf
23-09-2026 18:38          744 Fuse.class
23-09-2026 18:38          849 Fuse.java
23-09-2026 18:38    1,997 ImageFileManager.class
23-09-2026 18:38    1,997 ImageFileManager.java
23-09-2026 18:38           85 manifest.mf
23-09-2026 18:38    1,377 Pixelbenchmark.class
23-09-2026 18:38    1,318 Pixelbenchmark.java
23-09-2026 18:38    1,371 PrimarySets.class
23-09-2026 18:38    1,463 PrimarySets.java
23-09-2026 18:38    4,657 README.md
23-09-2026 18:38    779 README.md-
23-09-2026 18:38    83,769 RSA.class
23-09-2026 18:38    79,749 RSAnalysis.java
23-09-2026 18:38    5,208 RunStegExpose.class
23-09-2026 18:38    6,812 RunStegExpose.java
23-09-2026 18:38    8,822 SamplePairs.class
23-09-2026 18:38    3,418 SamplePairs.java
23-09-2026 18:38   10,621 stegExpose_analysis.pds
23-09-2026 18:38    5,666 stegExpose.jar
23-09-2026 18:38    756 testFolder.tlh
23-09-2026 18:38    testFeds
23-09-2026 18:38    25 File(s)      1,286,154 bytes
23-09-2026 18:38    2 Dir(s)      101,534,915,392 bytes free

C:\Users\Sarvegna\Downloads\StegExpose-master\StegExpose-master>jar -xvf StegExpose.jar testFolder
inflating: testFolder/
inflating: testFolder/amount of hidden data.txt 14778 bytes.
inflating: testFolder/amount of hidden data is 67141 bytes.

C:\Users\Sarvegna\Downloads\StegExpose-master\StegExpose-master>
```

5. Batch Analysis (Multiple Images)

- To check multiple images at once, specify the folder path containing the images:\

Command Structure:

java -jar StegExpose.jar <folder_path>

6. Advanced Options (Optional)

- To view additional parameters, such as options for adjusting detection sensitivity or output verbosity, use the --help flag:

Command:

```
File Edit View
File name, Above stego threshold?, Secret message size in bytes (ignore for clean files), Primary Sets, Chi Square, Sample Pairs, RS analysis, Fusion (mean)
clean_7221180570_822108c040_z.png, false, 13416, 0, 112292481888547, 0, 142350884449125, null, 0, 848735256189451834, 0, 10146154888774811
clean_7227262366_2688184c8e.png, false, 7391, NaN, 3, 741180013179327E-11, 0, 240066802558881854, 0, 25861310226721124, 0, 1662637596448026
clean_7228374852_8f743dc3d_z.png, false, 2449, 0, 0.01137468869920987, 0, 0.0073545822757812215, null, 0, 01607070550017578, 0, 01818758557850933
clean_7228718752_1c725eff7e_z.png, false, 14394, 0, 13616808612038, 0, 0.002182348817611224, null, 0, 0.0998112318120448, 0, 0.07780812429866454
clean_7231623538_f51db2a35a_z.png, false, 8198, 0, 0.090937672916859474, 0, 0.018913831823677696, null, 0, 0.06165268887585292, 0, 0.043834461872139026
clean_7231932608_c7f0b0ff25_z.png, false, 22569, 0, 0.2573883897346050, 0, 0.002651048945379063, 0, 0.09092183046233898, 0, 100751386937443394, 0, 11592787384593088
clean_7232202430_d38b6d6986_z.png, false, 25126, NaN, 0, 0.00698855773106866, 0, 1.8017057666795039, 0, 21813951225604777, 0, 133356048212415
clean_7232206610_bdcfdd20_z.png, false, 18190, 0, 0.02648841325646058, 0, 0.0396658763921852, 0, 0.000520823771301, 0, 0.08440531345457021, 0, 0.05734551904768359
clean_723220662_3d42c09109_z.png, false, 18000, NaN, 0, 0.01252686945037774, 0, 13287865993498356, 0, 1474074177378486, 0, 0.09760164904186998
clean_7234473332_10a4d48b2d.png, false, 1403, 0, 0.016399259187363, 0, 0.002616633877419855, 0, 0.014848472618129262
clean_7235558256_3099066753.png, false, 21450, 0, 1.7546102698982005, 0, 1.705494348209659, null, 0, 20167953078214375, 0, 18469366410764324
clean_7235972310_8c25288a5.png, false, 10882, 0, 18496458785274303, 0, 0.005156703935488528, null, 0, 0.0972843399055833, 0, 0.09580187725127161
clean_7237798856_86a78c8b23_z.png, false, 3054, 0, 0.07330206436492333, 0, 5.059594619348558E-11, 0, 0.0065819797476818, 0, 0.04793007393387279
stego_6666458261_8455d42625_z.png, true, 114785, NaN, 0, 0.06767021195235032, 0, 7292082212895838, 0, 7388125186819619, 0, 5116576586419387
stego_6672188499_85c582a7f9.png, true, 137847, NaN, 0, 0.0999085976238456, null, 0, 8608770381271956, 0, 93403138138752405
stego_6672542201_532f70bf7e.png, true, 67141, NaN, 0, 0.09371291967741853, 0, 7812415352997496, 0, 7680715936124942, 0, 54767673495298875
```

java -jar StegExpose.jar --help

7. Review the Results

- Review the scores generated for each image and use the threshold values to determine which images require further forensic investigation to extract the suspected hidden data.

Result:

The hidden data within the image was successfully detected using StegExpose. The analysis revealed that images with a suspect score above the threshold likely contain embedded steganographic content, confirming the tool's effectiveness in steganography detection.